Md. Jalish Mahamud

jalish.uftb@gmail.com
+8801749697937
scholar.google.com/Md. Jalish Mahamud
linkedin.com/Md Jalish Mahamud
github.com/jalish-uftb



Professional Summary

Internet of Things and Robotics Engineering graduate with hands-on experience in IoT, embedded systems, machine learning, and software development. Strong analytical thinker with experience in technical research, project development, and student leadership. Demonstrated ability to manage projects, coordinate teams, solve technical problems, and work effectively under deadlines. Seeking to contribute technical knowledge, problem-solving skills, and a results-oriented mindset.

Education

B.Sc. Engineering in Internet of Things and Robotics Engineering *2022 – 2026*
Institution: *University of Frontier Technology, Bangladesh*
Result: CGPA 3.45/4.00

Higher Secondary Certificate *2020*
Institution: *Govt. Tolaram College, Narayanganj*
Result: GPA 5.00/5.00
Board: Board of Intermediate and Secondary Education, Dhaka, Bangladesh

Secondary School Certificate *2018*
Institution: *Otra Secondary School, Wazirpur, Barishal*
Result: GPA 5.00/5.00
Board: Board of Intermediate and Secondary Education, Barishal, Bangladesh

Publications

Journal Papers

1. **Md. Jalish Mahamud**, Md. Sakibul Alam Sakib, Saurav Chandra Das, Fahmida A. Antara, and Sadia Enam. "Glove Z: A Wearable IoT and Machine Learning-Based Real-Time Health Monitoring System for Bedridden Patients." *Journal of Sensors*, *(Under Review)*, 2026. [Q2 Journal]

Conference Papers

1. Md. Numanur Rahman, Tahzib Mahmud Rifat, **Md. Jalish Mahamud**, and Suman Saha. "Explainable Deep Learning and IoT Fusion Strategy for Early Indoor Fire Safety Monitoring." *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, pp. 1–6, 2026. doi: 10.1109/QPAIN69676.2026.11546226.
2. **Md. Jalish Mahamud**, Md. Sakibul Alam Sakib, Rakib Hassan Dipu, Aar Raisatunnesa Hridika, and Saurav Chandra Das. "Hear2Learn: A Context-Aware Real-Time Captioning Assistant for Deaf and Hard-of-Hearing Students." *29th International Conference on Computer and Information Technology (ICCIT 2026). (Submitted).*

Research Interests

- Embedded Systems
- Quantum Computing
- Internet of Things (IoT)
- Robotics
- Human Machine Interface
- Explainable AI

Academic Projects

1. **Hear2Learn: Context-Aware Real-Time Captioning Assistant for DHH Students** *2026*
   Python, Flask, Flutter, Firebase, NLP, Speech Recognition
   - Developed a context-aware real-time captioning system to support deaf and hard-of-hearing students in classroom environments.
   - Implemented speech-to-text transcription with academic keyword extraction, examination-alert detection, announcements, and lecture summarization.
   - Designed a Flutter-based HMI with live transcripts, keyword highlighting, lecture history, and summary features.

2. **Glove Z: Wearable IoT and Machine Learning-Based Real-Time Health Monitoring System** *2025*
ESP32, MAX30102, DHT22, AD8232, Flex Sensors, Blynk, ThingSpeak, Flask, Python, Machine Learning, Twilio
- Developed a wearable IoT health monitoring prototype for bedridden patients using ESP32 with real-time heart rate, SpO_2, temperature, ECG, and gesture-based patient need detection.
- Integrated Blynk, ThingSpeak, and WhatsApp via Twilio for real-time visualization, cloud-based data logging, automatic alerts, and bidirectional caregiver communication.
- Implemented and evaluated six machine learning classifiers for Healthy, Warning, and Critical health-state classification, with a tuned Random Forest achieving 96.8% test accuracy on a 60,000-sample synthetic dataset.

3. **Advanced Farmhouse Security System** *2024*
Micropython, IoT, ExpressJS server, Embedded Systems, WebSocket, Database
- Designed and developed an IoT-based farmhouse security and environmental monitoring system using multiple sensors and microcontrollers.
- Implemented multimodal biometric authentication to provide secure user access and enhance overall system security.
- Integrated hardware, software, database, and data visualization components for real-time monitoring of environmental conditions, security events, and device status.

4. **LabAssist: A Comprehensive Laboratory Management System** *2024*
React, Node.js, Express.js, MongoDB, RESTful APIs
- Designed a laboratory management system to reduce paperwork and streamline routine administrative processes.
- Developed workflows for equipment requests, approval management, user authentication, and laboratory data management.
- Structured role-based access for administrators, laboratory technicians, and students or researchers to support organized laboratory operations.

5. **AI-Based Human Following Robot** *2023*
Artificial Intelligence, Computer Vision, Robotics, Embedded Systems
- Developed an intelligent robotic system designed to detect and track a human target and autonomously follow their movement.
- Integrated human-tracking logic with robotic motion control to support real-time target following and distance maintenance.
- Explored obstacle detection and avoidance to improve the robot's navigation and operational safety.

Bachelor Thesis

Title: "CROSS-XFed: Cross-Dataset Federated and Explainable Intrusion Detection for Heterogeneous IoT Networks"

Achievements

- **2026: Dean's List**, Level 4 Term 1 & Term 2, University of Frontier Technology, Bangladesh.

Research and Technical Skills

- **AI & Machine Learning:** Deep Learning, Scikit-learn,Flask ML Deployment, Model Evaluation, Explainable AI (XAI)
- **Programming Languages:** Python, C, C++, Java, SQL (MySQL)
- **IoT & Embedded Systems:** Microcontrollers (ESP32/Arduino), Sensor Interfacing, Hardware-Software Integration, WebSockets, Real-Time Data Streaming
- **Web & Mobile Development:** HTML, CSS, Flutter, RESTful APIs, Node.js
- **Research Tools:** LaTeX, Git & GitHub, VS Code, MS Office tools

Language Skills

Bengali	Mother tongue
English	Excellent proficiency

Professional Development and Certifications

Industry-Standard PCB Schematic and Layout Design *2025*
UFTB Career Center

Foundation of Python *2024*
Ostad

Cyber Security Training Program *2023*
Creative IT Institute

Job Ready Certification: 21st Century Core Employability Skills Program *2023*
Wadhwani Foundation

Leadership Experience

1. **President — Barishal Divisional Students Welfare Association (BDSWA), UFTB** *2024 – 2026*
   - Led the planning and execution of student programs and activities.
   - Coordinated volunteers and managed event responsibilities while strengthening teamwork.
2. **Vice President — UFTB Robotics Club** *2025 – 2026*
   - Organized technical workshops and supported project coordination among club members.
3. **Executive Member — UFTB STEM Club** *2023 – 2024*
   - Contributed to STEM-related initiatives and academic events.

Personal Details

Date of Birth	June 02, 2003
Gender	Male
Nationality	Bangladeshi
Blood Group	A(+ve)
Present Address	Latifpur, Kaliakair, Gazipur-1750, Bangladesh
Permanent Address	Otra, Wazirpur, Barishal, Bangladesh

References

Md. Ashiqussalehin
Lecturer
Department of Internet of Things and Robotics Engineering
University of Frontier Technology, Bangladesh
Email: ashiqussalehin0001@uftb.ac.bd
Phone: +8801799776562

Md. Toukir Ahmed
Assistant Professor
Department of Internet of Things and Robotics Engineering
University of Frontier Technology, Bangladesh
Email: toukir0001@uftb.ac.bd
Phone: +8801738039128